



STN « 2025 » _ « 2026 »

Développement
d'un système
embarqué pour
mesurer et
estimer la VO₂
max à partir des
tests de Léger et
Cooper.

Projet : STN21 – Système embarqué VO₂max

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Code Émetteur – ESP32 (Test de Léger)

This program implements the embedded system of the Léger test using an ESP32 Heltec LoRa V3 board. It measures distance using an ultrasonic sensor, generates audio signals to control the running pace, detects runner passages, and calculates $VO_2\text{max}$. The data is transmitted in real time using LoRa communication.

Code (commented) :

```
#include <Wire.h>

#include <SSD1306Wire.h>

#include <RadioLib.h>

#include "pins_arduino.h"

// ===== OLED =====

// Initialize OLED display (I2C)
SSD1306Wire display(0x3C, SDA_OLED, SCL_OLED);

// Enable OLED power
void VextON() {
  pinMode(Vext, OUTPUT);
  digitalWrite(Vext, LOW); // LOW = ON
}

// Reset OLED display
void displayReset() {
  pinMode(RST_OLED, OUTPUT);
  digitalWrite(RST_OLED, HIGH);
  delay(50);
  digitalWrite(RST_OLED, LOW);
  delay(50);
  digitalWrite(RST_OLED, HIGH);
  delay(50);
}

// ===== LoRa =====
```

```
#define LORA_FREQ 868.0
```

```
// Initialize LoRa module
```

```
SX1262 radio = new Module(SS, DIO0, RST_LoRa, BUSY_LoRa);
```

```
bool loraOK = false;
```

```
// Setup LoRa communication
```

```
void setupLoRa() {
```

```
    int state = radio.begin(LORA_FREQ);
```

```
    if (state != RADIOLIB_ERR_NONE) {
```

```
        loraOK = false;
```

```
        return;
```

```
    }
```

```
// LoRa parameters configuration
```

```
radio.setSpreadingFactor(9);
```

```
radio.setBandwidth(125.0);
```

```
radio.setCodingRate(7);
```

```
radio.setOutputPower(14);
```

```
radio.setSyncWord(0x12);
```

```
loraOK = true;
```

```
}
```

```
// ===== TEST DE LEGER =====
```

```
// Pin configuration
```

```
const int TRIG_PIN = 45;
```

```
const int ECHO_PIN = 46;
```

```
const int BUZZER_PIN = 6;
```

```
const int BUTTON_PIN = 4;
```

```
// Detection threshold
```

```
const int SEUIL_CM = 150;
```

```
// Test parameters

#define N_PALIERS 20

const float V_DEBUT = 8.5;

const float V_INC  = 0.5;


// Number of shuttles per level

const int NAVETTES_PAR_PALIER[N_PALIERS] =
{7,8,8,9,9,10,10,10,11,11,11,11,12,12,13,13,13,14,14,15};


// Variables

int palier = 1;

int navette = 0;

float vitesse_kmh = V_DEBUT;


// ===== FUNCTIONS =====


// Measure distance using ultrasonic sensor
long mesureDistanceCm() {
    digitalWrite(TRIG_PIN, LOW);
    delayMicroseconds(3);
    digitalWrite(TRIG_PIN, HIGH);
    delayMicroseconds(10);
    digitalWrite(TRIG_PIN, LOW);
    long duree = pulseIn(ECHO_PIN, HIGH, 30000UL);
    if (duree == 0) return -1;
    return (long)(duree * 0.034 / 2.0);
}


// Generate short beep

void bipCourt() {
    digitalWrite(BUZZER_PIN, LOW);
```

```
    delay(80);
    digitalWrite(BUZZER_PIN, HIGH);
}

// Generate long beep
void bipLong() {
    digitalWrite(BUZZER_PIN, LOW);
    delay(250);
    digitalWrite(BUZZER_PIN, HIGH);
}

// End of test and VO2max calculation
void finTest() {
    float vo2max = 3.5 * vitesse_kmh;

    // Send result via LoRa
    if (loraOK) {
        String msg = "V:" + String(vo2max, 1);
        radio.transmit(msg);
    }
}

// ===== SETUP =====
void setup() {
    Serial.begin(115200);

    VextON();
    displayReset();
    display.init();

    pinMode(TRIG_PIN, OUTPUT);
    pinMode(ECHO_PIN, INPUT);
```

```
pinMode(BUZZER_PIN, OUTPUT);  
pinMode(BUTTON_PIN, INPUT_PULLUP);
```

```
setupLoRa();  
}
```

```
// ===== LOOP =====
```

```
void loop() {
```

```
    // Wait for button press to start test
```

```
    if (digitalRead(BUTTON_PIN) == LOW) {
```

```
        // Measure distance continuously
```

```
        long distance = measureDistanceCm();
```

```
        // Detect runner passage
```

```
        if (distance > 0 && distance < SEUIL_CM) {
```

```
            bipCourt();
```

```
        }
```

```
        // Additional logic for timing and levels here...
```

```
    }
```

```
}
```

Code Récepteur – ESP32 (LoRa)

This program implements a LoRa receiver using an ESP32 Heltec board. It receives data transmitted by the emitter and displays it on an OLED screen in real time.

Code (commented) :

```
#include <Wire.h>
```

```
#include <SSD1306Wire.h>
```

```
#include <RadioLib.h>
```

```
#include "pins_arduino.h"
```

```
// OLED initialization
```

```
SSD1306Wire display(0x3C, SDA_OLED, SCL_OLED);
```

```
// Enable OLED power
```

```
void VextON() {
```

```
    pinMode(Vext, OUTPUT);
```

```
    digitalWrite(Vext, LOW);
```

```
}
```

```
// Reset display
```

```
void displayReset() {
```

```
    pinMode(RST_OLED, OUTPUT);
```

```
    digitalWrite(RST_OLED, HIGH);
```

```
    delay(50);
```

```
    digitalWrite(RST_OLED, LOW);
```

```
    delay(50);
```

```
    digitalWrite(RST_OLED, HIGH);
```

```
}
```

```
// LoRa configuration
```

```
#define LORA_FREQ 868.0
```

```
SX1262 radio = new Module(SS, DIO0, RST_LoRa, BUSY_LoRa);
```

```
// ===== SETUP =====  
  
void setup() {  
  Serial.begin(115200);  
  
  VextON();  
  displayReset();  
  display.init();  
  
  // Initialize LoRa receiver  
  int state = radio.begin(LORA_FREQ);  
  
  // Configure LoRa parameters  
  radio.setSpreadingFactor(9);  
  radio.setBandwidth(125.0);  
  radio.setCodingRate(7);  
  radio.setSyncWord(0x12);  
}  
  
// ===== LOOP =====  
  
void loop() {  
  String msg;  
  
  // Receive LoRa message  
  int state = radio.receive(msg);  
  
  // If message received correctly  
  if (state == RADIOLIB_ERR_NONE) {  
    display.clear();  
    display.drawString(0, 0, msg);  
    display.display();  
  }  
}
```


}

Projet	STN21 – Système embarqué VO2 max (Test Léger-Test Cooper)
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